

Python Practice Problems

Topic: if-elif-else and for loops (with hints)

If-elif-else

- 1 Take a number as input and check whether it is positive, negative, or zero.
Hint: Use `if num > 0, elif num < 0, else`.
- 2 Take a number as input and check whether it is even or odd.
Hint: Check `num % 2 == 0`.
- 3 Take three numbers as input and print the largest among them.
Hint: Compare using `if a >= b and a >= c`, then `elif`, `else`.
- 4 Take a student's marks as input and print the grade (A, B, C, D, F) based on ranges.
Hint: Check from highest range down using `elif`.
- 5 Take age as input and check whether the person is eligible to vote (18 or above).
Hint: Simple check `if age >= 18`.
- 6 Take a year as input and check whether it is a leap year or not.
Hint: Divisible by 4 but not 100, unless also by 400.
- 7 Take a character as input and check whether it is a vowel or consonant.
Hint: Use `if ch.lower() in 'aeiou'`.
- 8 Take a number as input and check whether it is divisible by both 3 and 5, only 3, only 5, or neither.
Hint: Check the "both" condition first with `and`.
- 9 Take the price of an item as input and calculate the discount based on price range.
Hint: Use `elif` from highest price range downward.
- 10 Take three sides of a triangle as input and check validity and type (equilateral, isosceles, scalene).
Hint: Sum of any two sides > third side; then compare side equality.

For Loops

- 1 Print numbers from 1 to 10 using a for loop.
Hint: `for i in range(1, 11):`
- 2 Print the multiplication table of a number entered by the user.
Hint: `for i in range(1, 11): print(num * i)`
- 3 Calculate the sum of first n natural numbers using a for loop.
Hint: Use a `total = 0` variable and add `i` each loop.
- 4 Calculate the factorial of a number using a for loop.
Hint: Use `fact = 1` and multiply by `i` each loop.
- 5 Print all even numbers between 1 and 50.
Hint: Use `range(2, 51, 2)`.
- 6 Count the number of digits in a given number using a for loop.
Hint: Convert number to string and use `len()`.
- 7 Print the Fibonacci series up to n terms.
Hint: Start with `a, b = 0, 1`, update `a, b = b, a+b`.
- 8 Take a number as input and check whether it is prime using a for loop.
Hint: Check if any number from 2 to `num-1` divides it evenly.
- 9 Print a pattern of stars in the shape of a right-angled triangle using nested for loops.
Hint: Outer loop for rows, inner loop for stars, use `end=""`.
- 10 Reverse a given number using a for loop.
Hint: Use `num % 10` and `num // 10` to extract and rebuild digits.